Cluster Analysis

Readings: Zelterman, 2015, Chapter 11; Izenman, 2008, Chapters 12.1–12.4, 12.9; ISLR, 2021, Chapter 12.4; Johnson & Wichern, 2007, Chapters 12.1–12.5

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Agenda

Overview

K-Means Clustering

Hierarchical Clustering

Model-Based Clustering

Learning Objectives

By the end of this lecture, you should be able to:

- **Explain the goal of clustering** and distinguish it from supervised classification
- **Apply K-means clustering** and explain the roles of distance measures, variable scaling, centroids, and the number of clusters
- **Describe hierarchical clustering** and interpret dendrograms and linkage methods.
- **Explain model-based clustering** using Gaussian mixture models, and evaluate, compare, and interpret clustering results across methods

What is Cluster Analysis?

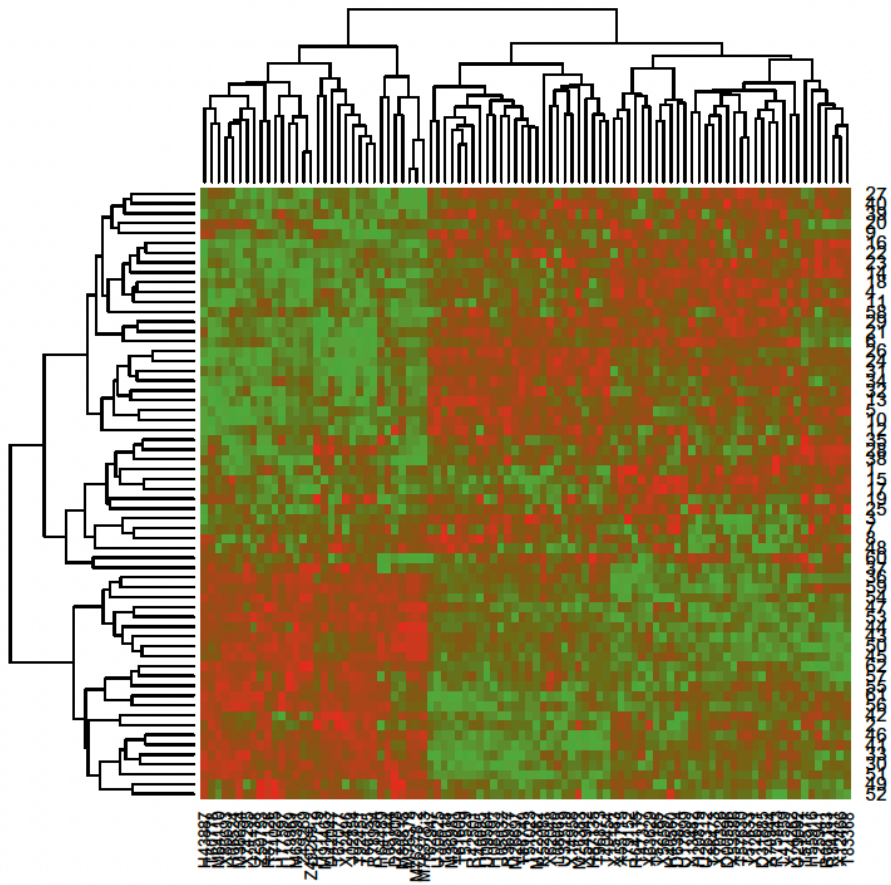
- Cluster analysis groups observations based on their **similarity**, without predefined class labels
- The goal is to identify meaningful patterns or structures in the data
- Unlike supervised classification, neither the cluster labels nor the number of groups is generally known

Key question: What makes two observations similar?

Some Examples of Clustering Applications

- Market segmentation: Help marketers discover distinct groups in their customer bases and use this knowledge to develop targeted marketing programs.

- Clustering gene expression data:



What Is Good Clustering?

- High within-cluster similarity and low between-cluster similarity
- Similarity depends on the **distance measure**, **selected variables**, and their **scales**
- A useful clustering should reveal interpretable and meaningful structure

Example: When clustering customers using income and satisfaction scores, standardization may be necessary to prevent income from dominating Euclidean distance.

Major Clustering Approaches

- **Partitioning algorithm:** Partition the observations into a prespecified number of clusters; for example, K-means clustering.
- **Hierarchical algorithm:** Construct a hierarchical decomposition of the observations; for example, hierarchical agglomerative clustering.
- **Model-based clustering:** Specify a probabilistic model for each cluster; for example, Gaussian mixture models.

Partitioning Clustering

- Partitioning methods divide n observations into K **non-overlapping clusters**
- Let $C_1, \ldots, C_K$ denote the sets of observation indices assigned to each cluster
- A valid partition satisfies two conditions:
  - **Complete assignment:** Every observation belongs to a cluster.

$$\bigcup_{k=1}^{K} C_k = \{1, \ldots, n\}.$$

  - **No overlap:** Each observation belongs to only one cluster.

$$C_k \cap C_{k'} = \emptyset \quad \text{for } k \neq k'$$

Example: If observation $\boldsymbol{x}_i$ is assigned to cluster k, then $i \in C_k$.

The K-Means Algorithm

1. **Choose K:** Specify the number of clusters.

2. **Initialize:** Select K initial cluster centroids (e.g., randomly chosen observations).

3. **Repeat until assignments stabilize:**

    - **Assignment step:** Assign each observation to its nearest centroid using Euclidean distance.

    - **Update step:** Recompute each centroid as the mean of the observations assigned to that cluster.

Assign observations $\longrightarrow$ Update centroids $\longrightarrow$ Repeat

What Does K-Means Optimize?

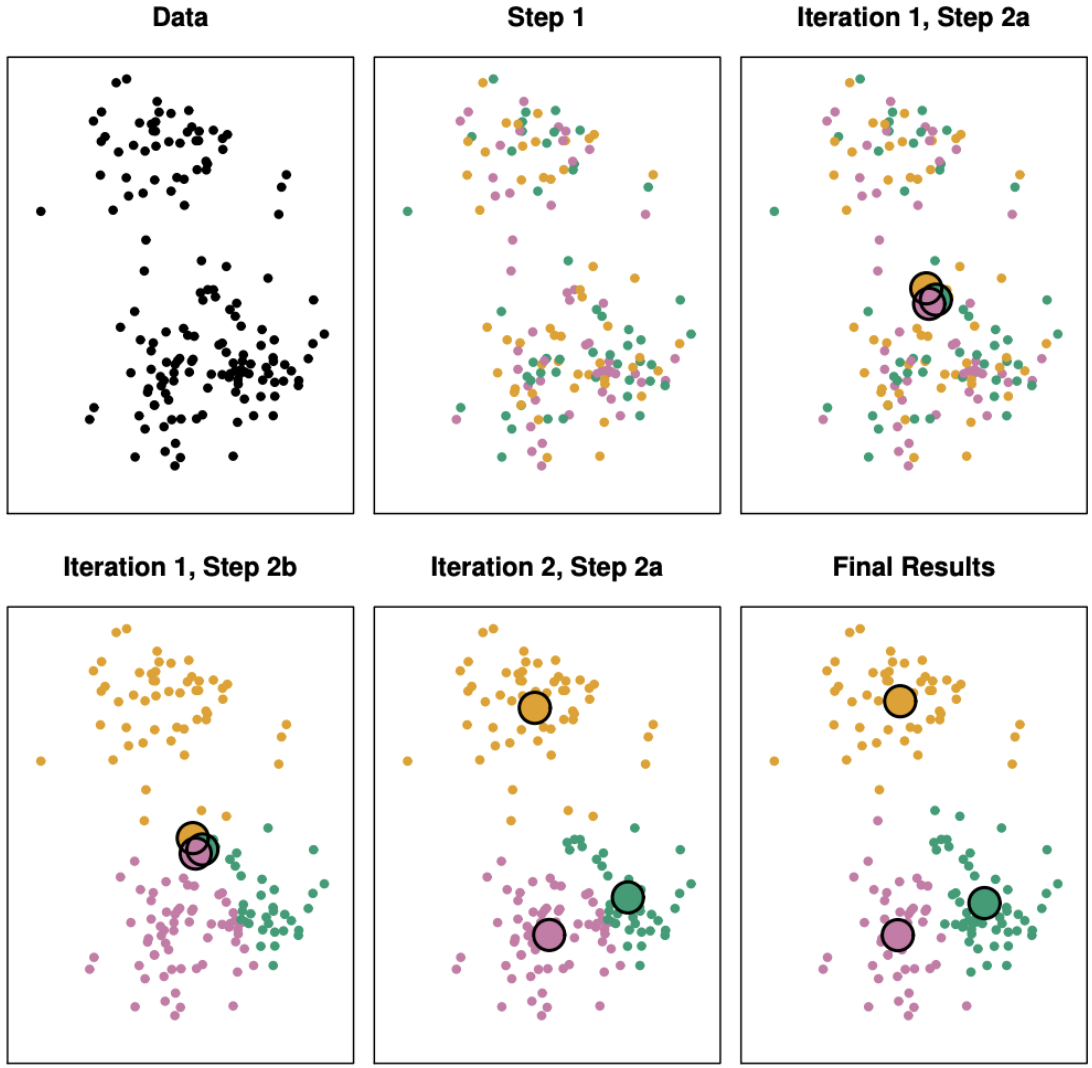
- K-means divides observations into K clusters by minimizing the **within-cluster sum of squares (WCSS)**.

$$\min_{C_1,\ldots,C_K} \quad W = \sum_{k=1}^{K} \sum_{i \in C_k} \|\boldsymbol{x}_i - \boldsymbol{\mu}_k\|^2$$

- $\boldsymbol{\mu}_k = \dfrac{1}{|C_k|} \displaystyle\sum_{i \in C_k} \boldsymbol{x}_i$: the **centroid** (mean) of cluster k.

Key idea: Find clusters whose observations are as close as possible to their own centroids.

K-Means Clustering Illustration



Assign $\rightarrow$ **Update** $\rightarrow$ **Repeat**: Each iteration improves or maintains the clustering objective

How Many Clusters?

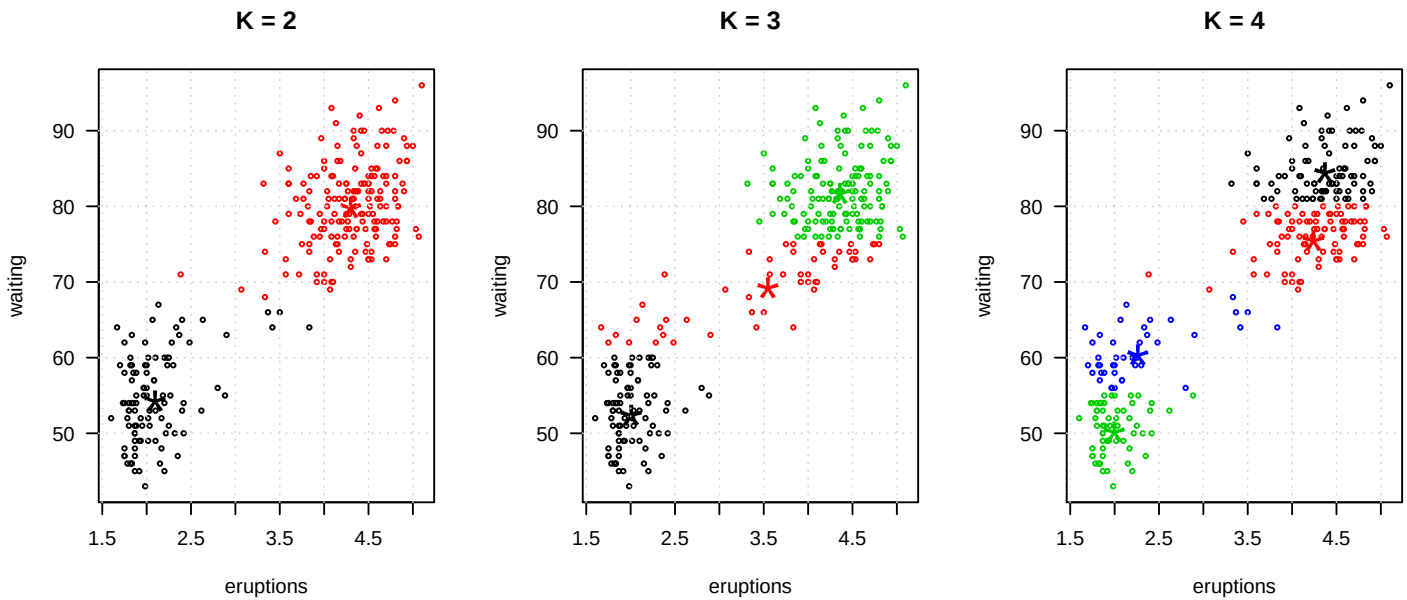
- K-means requires us to specify the number of clusters, K, in advance
- The **within-cluster sum of squares (WCSS)** decreases (or stays unchanged) as K increases
  - More clusters always improve the fit, but may not produce more meaningful groups
- Two common approaches for choosing K:
  - **Elbow method:** Look for where the reduction in WCSS begins to level off
  - **Silhouette analysis:** Favor clusters that are compact and well separated

Key Takeaway: Choose K to balance cluster quality, interpretability, and application goals

K-Means Clustering in R

```
kmean3.faithful <- kmeans(x = faithful, centers = 3)
```

- `kmeans(x, centers = K, nstart = 25)`: Run K-means with 25 random initializations and retain the solution with the smallest WCSS
- **Compare** $K = 2, 3, 4$**:** How do cluster separation, sizes, and interpretation change?



Hierarchical Clustering

- **K-means** produces a single partition of the data into a prespecified number of clusters, K
- **Hierarchical clustering** builds a sequence of **nested clusters**, allowing us to explore different numbers of groups without choosing K in advance
- In **agglomerative clustering** (a "bottom-up" approach):
  - Each observation initially forms its own cluster
  - At each step, the two closest clusters are merged, gradually forming larger groups

Key Difference: K-means finds one partition; hierarchical clustering reveals a hierarchy of possible partitions

Hierarchical Clustering Algorithm

Agglomerative (bottom-up) clustering:

1. **Initialize:** Start with n clusters, one for each observation
2. **Find the closest pair:** Identify the two clusters with the smallest inter-cluster dissimilarity
3. **Merge:** Combine these two clusters into one.
4. **Repeat:** Update the inter-cluster dissimilarities and repeat until all observations form a single cluster

Key Idea: Build a hierarchy by repeatedly merging the closest clusters

How Do We Measure Distance Between Clusters?

The **linkage criterion** determines how the dissimilarity between two clusters is measured.

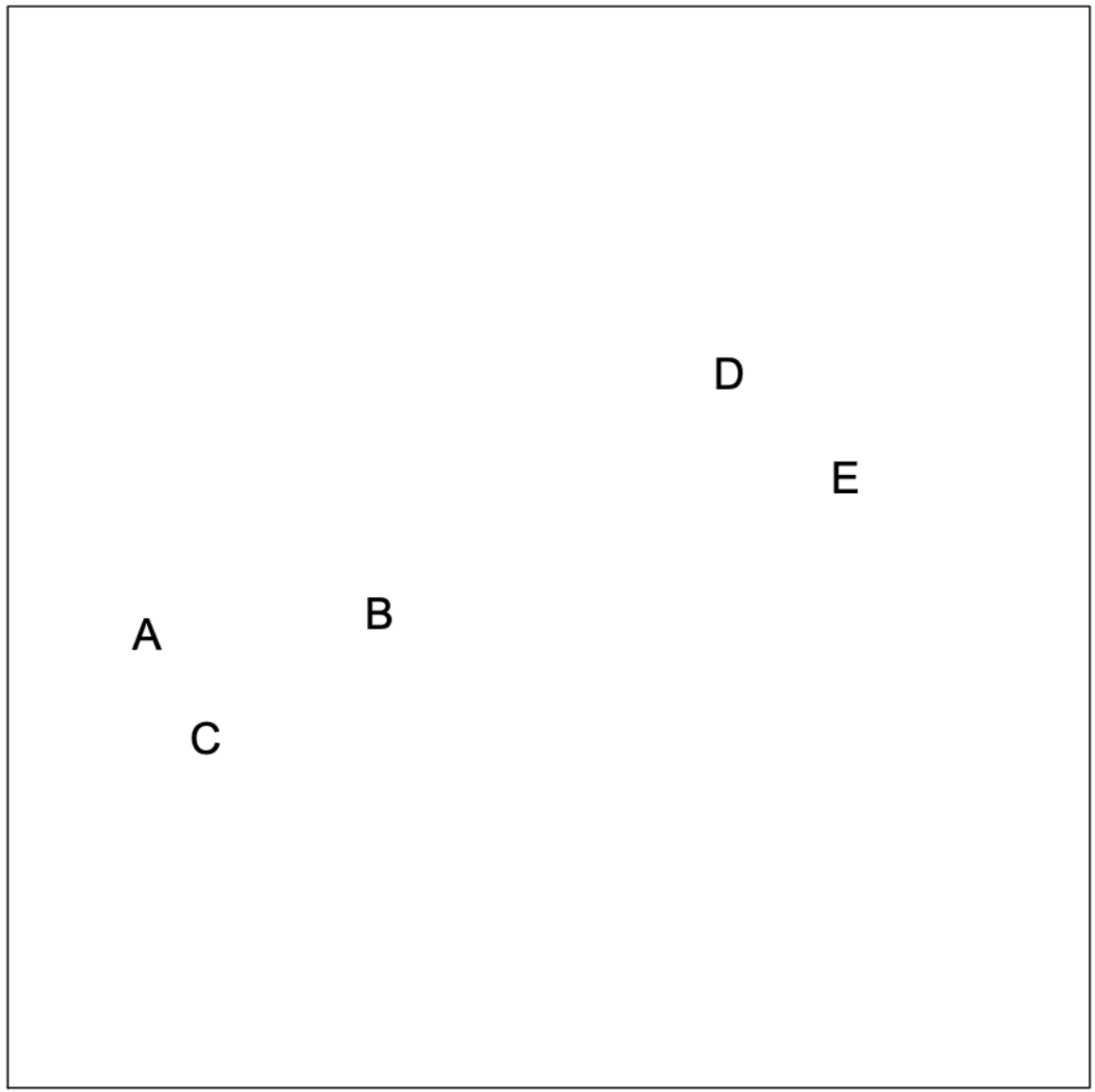
Linkage Method	**Criterion**
Single	Minimum pairwise distance
Complete	Maximum pairwise distance
Average	Average pairwise distance
Ward	Minimum increase in within-cluster SS

- **Single:** Can form long, chain-like clusters.
- **Complete / Ward:** Often favor compact clusters.

Key Takeaway: Different linkage criteria can produce different clustering hierarchies.

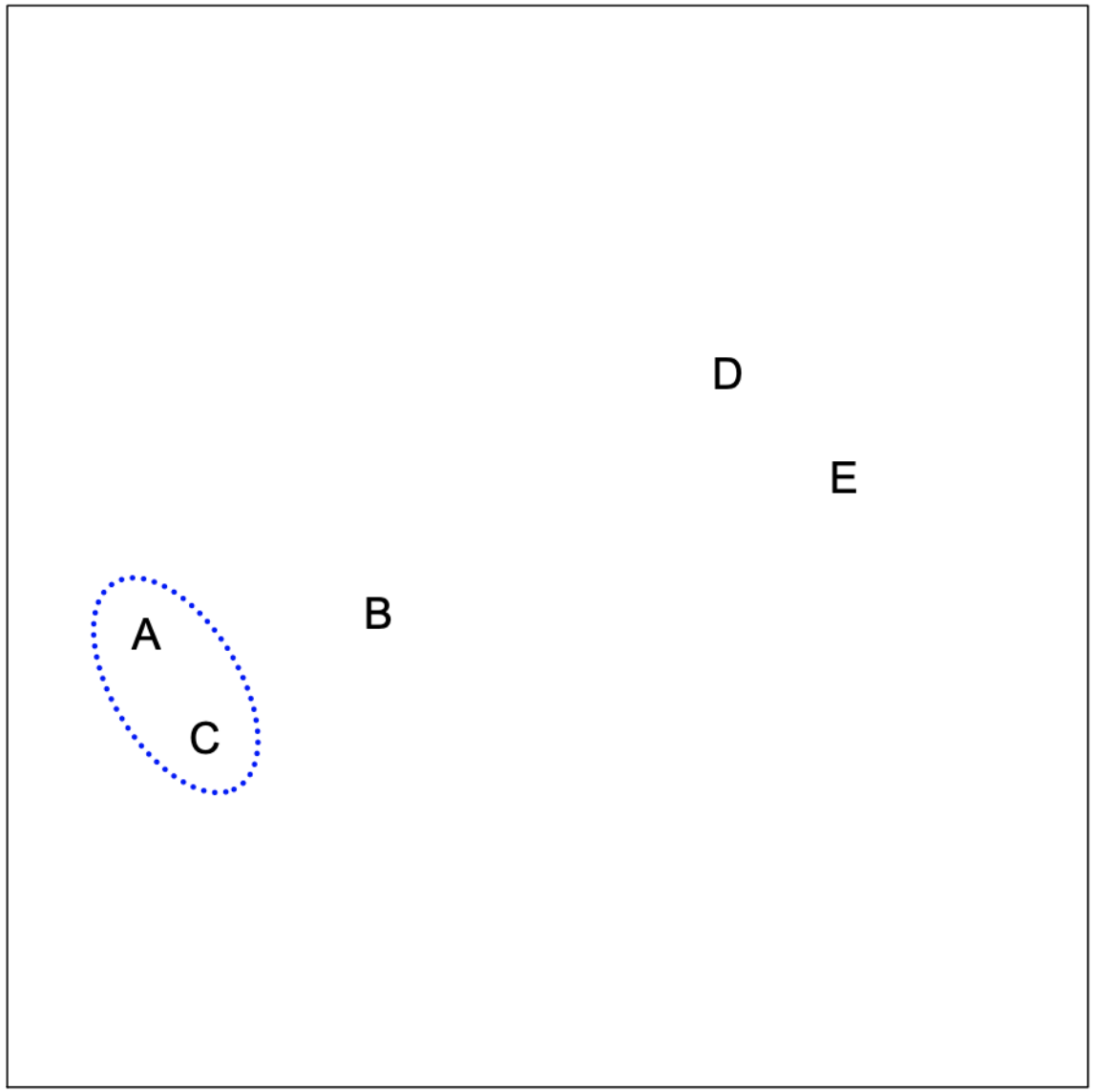
Hierarchical Clustering: Starting Point

Step 1: Each observation starts as its own cluster.



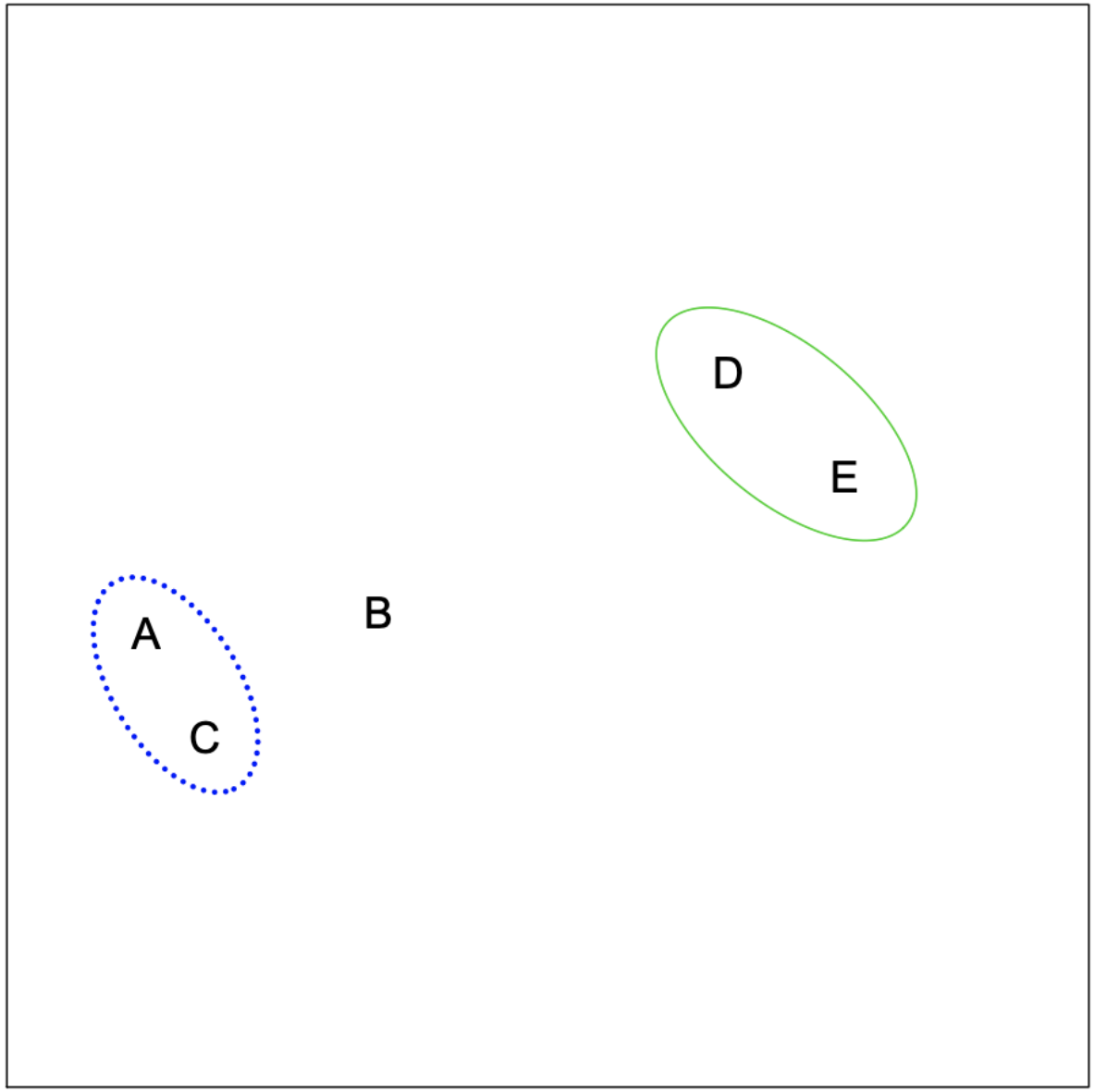
Hierarchical Clustering: First Merge

Step 2: Merge the two closest clusters.



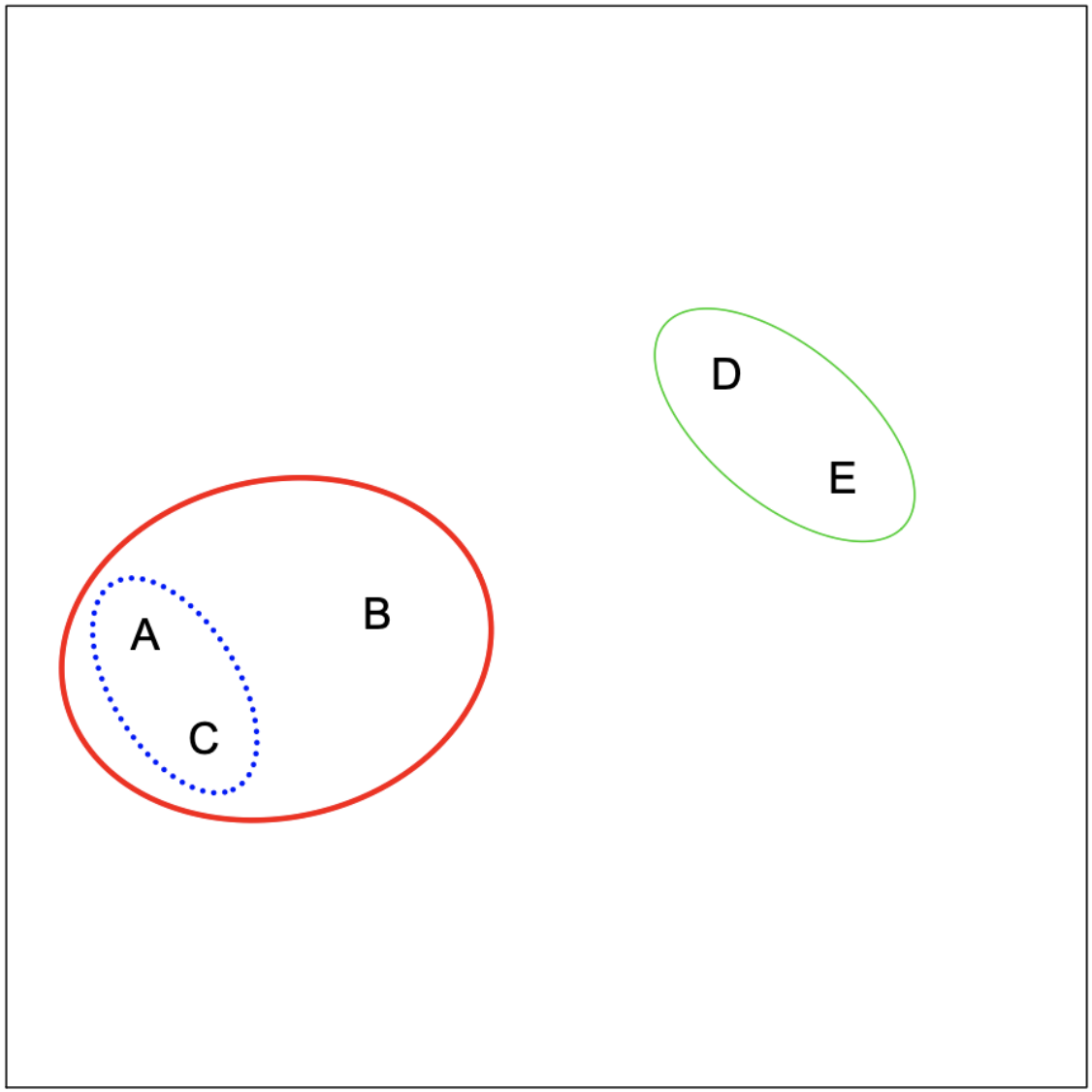
Hierarchical Clustering: Updating Distances

Step 3: Recompute distances between clusters.



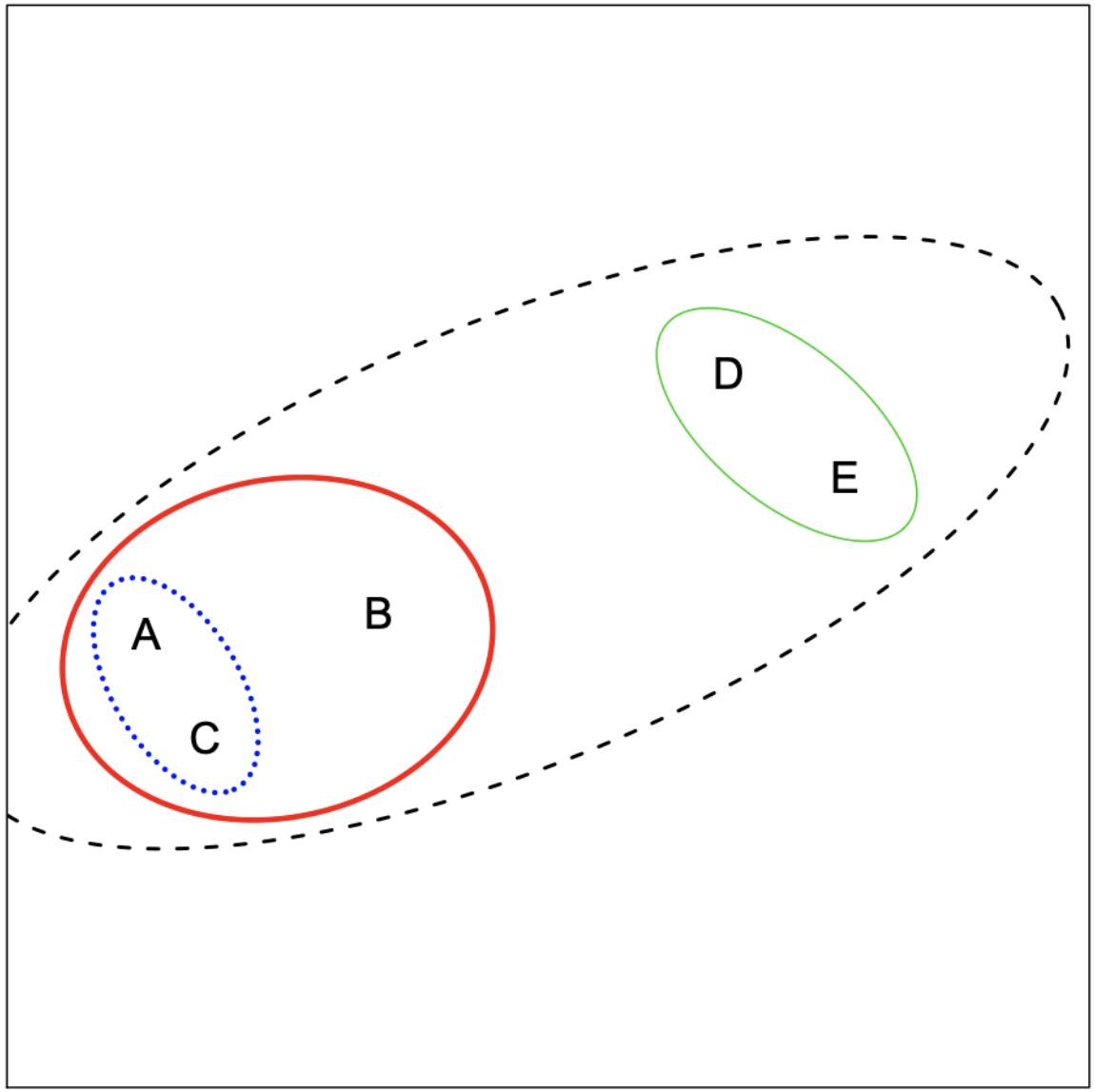
Hierarchical Clustering: Building the Hierarchy

Step 4: Continue merging the closest clusters.



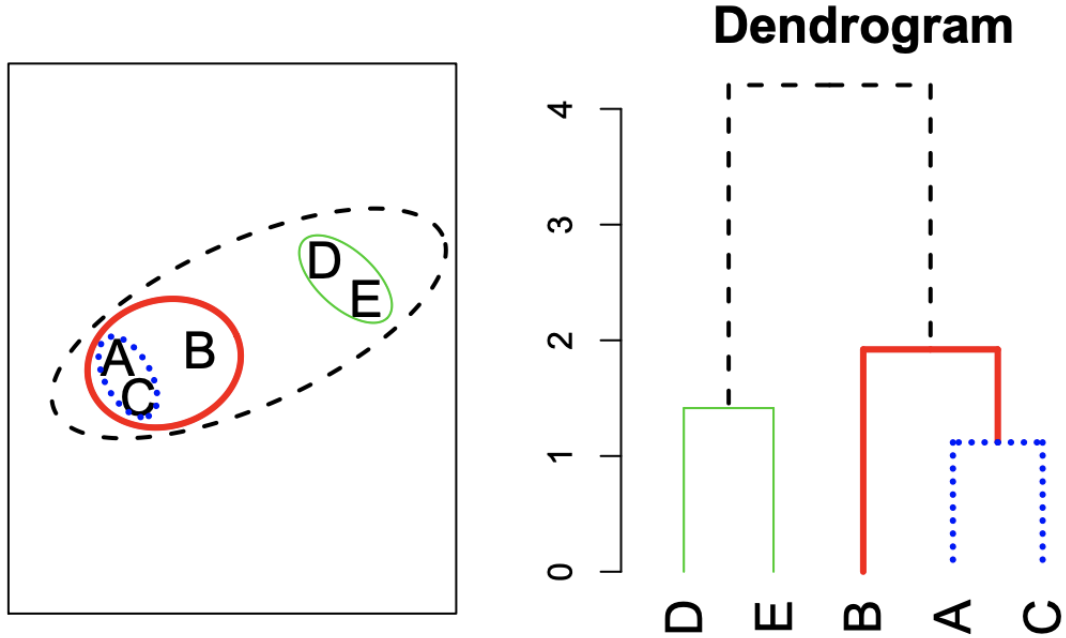
Hierarchical Clustering: The Final Hierarchy

Continue until all observations form one cluster.



How to Read a Dendrogram

- Each branch joining two clusters represents a **merge**
- The **merge height** reflects the dissimilarity (or criterion) used by the selected linkage method
- Cutting the dendrogram at a chosen height produces a **partition** of the observations

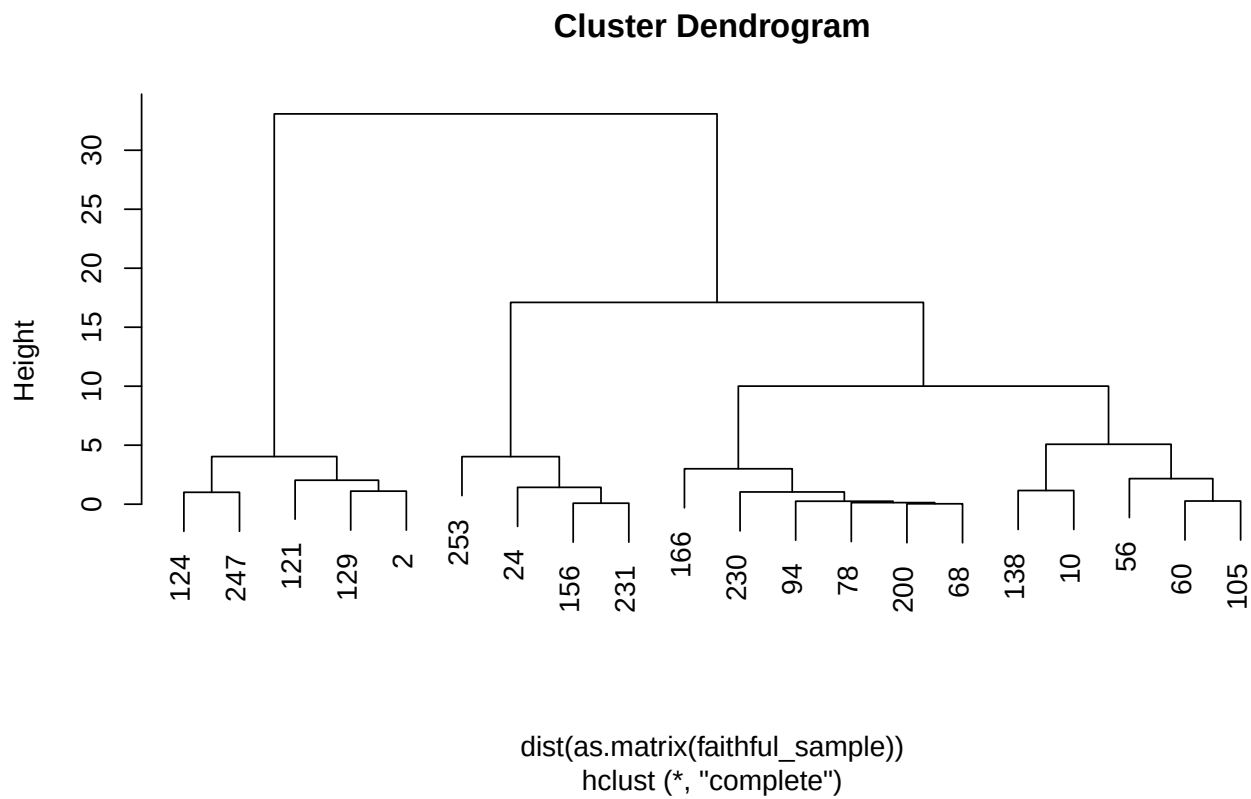


Key Takeaway: A dendrogram displays a hierarchy of possible clusterings, not just one fixed partition

Hierarchical Clustering in R

```
hc.faithful <- hclust(dist(faithful_sample))
plot(hc.faithful)
```

- `hclust()`: Builds the hierarchy (**complete linkage** by default).
- `cutree(hc.faithful, k = 3)`: Assigns observations to **three clusters**.



Model-Based Clustering

- **K-means** groups observations by minimizing within-cluster variation, without specifying a probability model for the data
- **Model-based clustering** assumes observations arise from a mixture of probability distributions
- This probabilistic framework allows us to:
  - Model cluster shapes and variability
  - Estimate probabilities of cluster membership
  - Compare models using statistical criteria

Key Idea: Model-based clustering describes both cluster structure and membership uncertainty

Gaussian Mixture Models

Assume observations arise from a mixture of K multivariate Gaussian distributions:

$$f(\boldsymbol{x}) = \sum_{k=1}^{K} \pi_k \, \phi_p(\boldsymbol{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

Each component (cluster) has:

- π_k: **Mixing proportion** ($\pi_k > 0$, $\sum_k \pi_k = 1$).
- $\boldsymbol{\mu}_k$: **Mean vector** (cluster center).
- $\boldsymbol{\Sigma}_k$: **Covariance matrix** (cluster shape and spread).

Key Idea: Estimate the mixture parameters and use them to calculate cluster membership probabilities

Hard vs. Probabilistic Clustering

- **K-means (hard clustering):** Each observation is assigned to exactly one cluster.
- **Gaussian mixtures (probabilistic):** Each observation receives estimated membership probabilities across clusters.

Example: Membership in Three Clusters

	Cluster 1	Cluster 2	Cluster 3
Observation A	0.95	0.03	0.02
Observation B	0.45	0.50	0.05

Observation A has a clear assignment; Observation B has greater membership uncertainty

Key Takeaway: Probabilistic clustering quantifies membership uncertainty under the fitted model

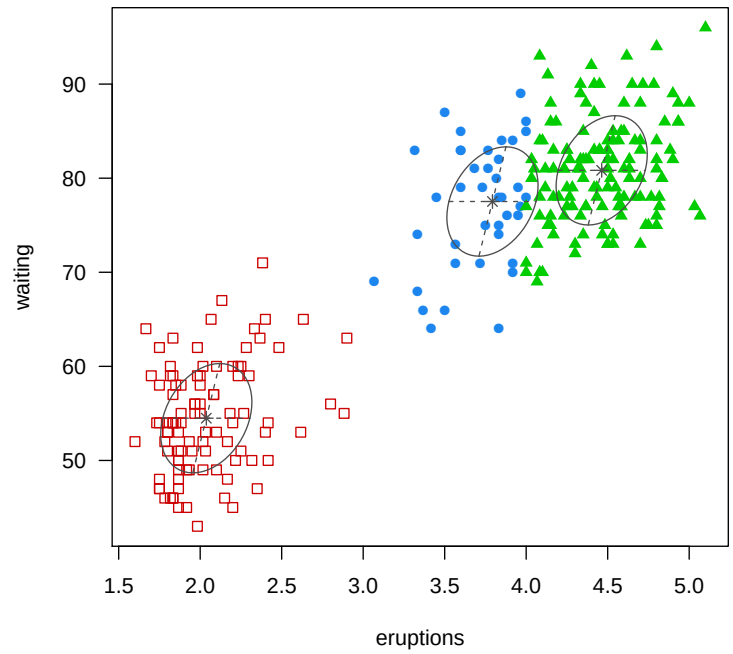
Gaussian Mixture Models in R

```
library(mclust)
```

```
## Package 'mclust' version 5.4.5
## Type 'citation("mclust")' for citing this R package in publications.
```

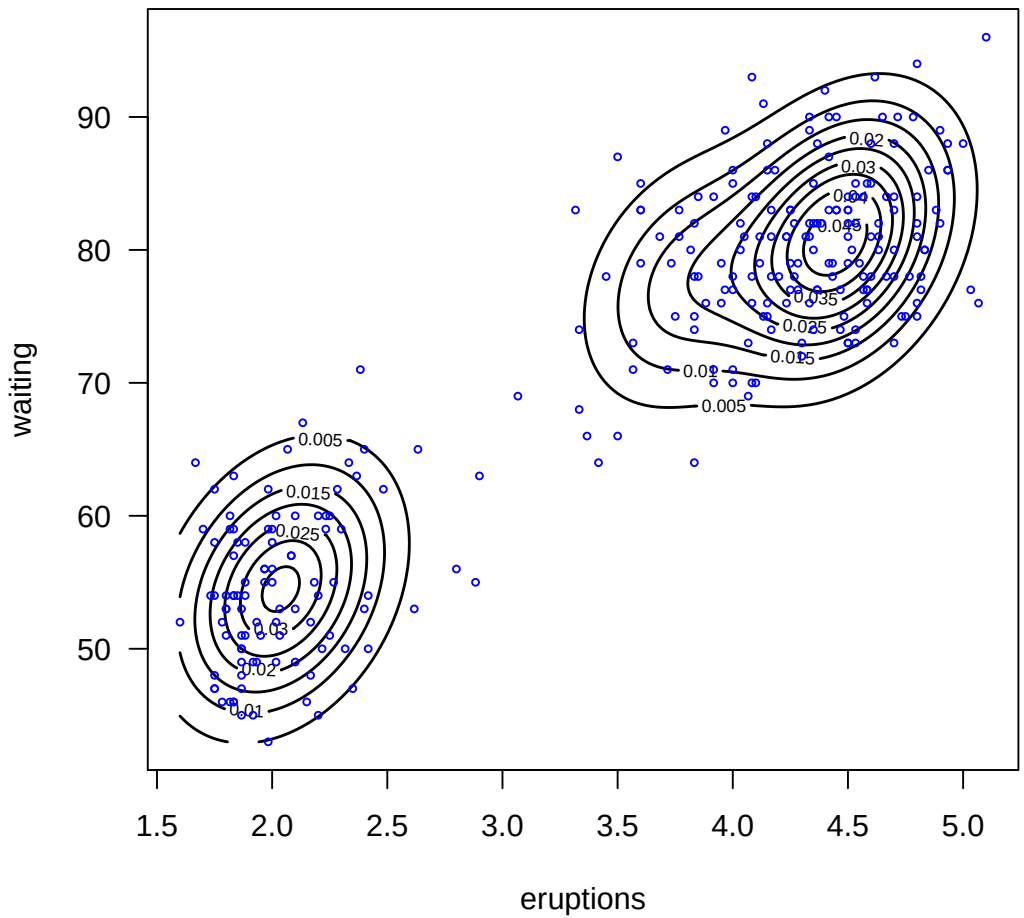
```
BIC <- mclustBIC(faithful)
model1 <- Mclust(faithful, x = BIC)
```

- `Mclust()` fits Gaussian mixture models and selects a model using **BIC**.
- **BIC** balances model fit and complexity across numbers of components and covariance structures.

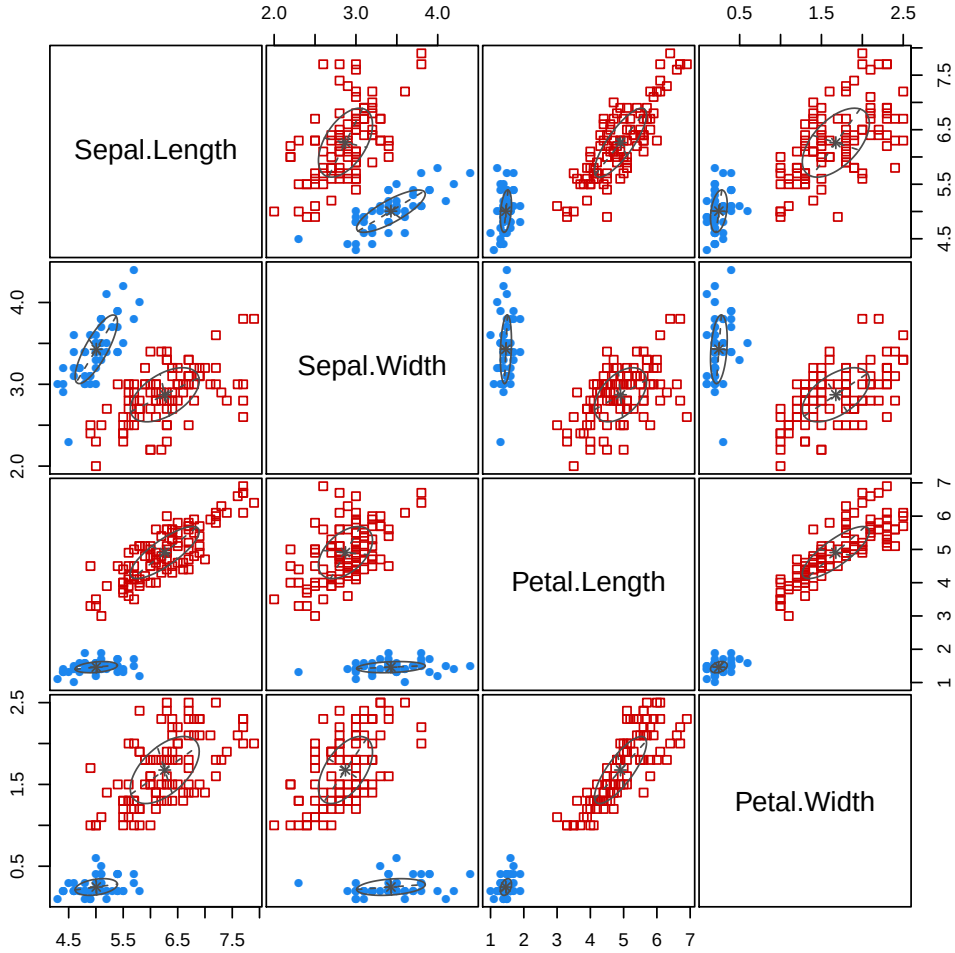


Interpreting Gaussian Mixture Components

- **Mean vectors:** Determine the centers of Gaussian components
- **Covariance matrices:** Determine their spread, shape, and orientation
- **Overlapping components:** May indicate uncertainty in cluster membership



Model-Based Clustering: Iris Data



Key Takeaway: Clusters are estimated **without species labels**; known labels can be used afterward to evaluate the clustering

How Do We Know Whether Clustering Is Useful?

- **Cluster quality:** Are observations within clusters similar and different clusters well separated?
  - Example: Silhouette score.
- **Stability:** Do similar clusters emerge under different initializations or reasonable data perturbations?
- **Practical interpretation:** Do the clusters reveal meaningful or actionable differences?

Key Takeaway: Finding clusters does not necessarily mean that meaningful cluster structure exists

Choosing a Clustering Method

Feature	**K-means**	**Hierarchical**	**Gaussian Mixture**
Principle	Minimize within-cluster SS	Merge using linkage	Fit mixture density
Choose K first?	Yes	No	Not necessarily
Membership	Hard	Hard after cutting	Probabilistic
Cluster shapes	Favors compact groups	Linkage-dependent	Covariance-dependent
Key consideration	Initialization, K	Distance, linkage	Model assumptions

Key Takeaway:

There is no universally best clustering method. Choose based on data structure, assumptions, and analytical goals.

Next lecture: Multidimensional Scaling